# PROJECT REPORT PART B

GROUP 44

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## Title of Project

- Simulation and Analysis of JFET Output Characteristics.

# Objective of Project

- To simulate, observe, and analyze the output characteristics of a Junction Field-Effect Transistor (JFET), infer its behavior, and explain why it functions as a voltage-controlled current source and possesses high input impedance.



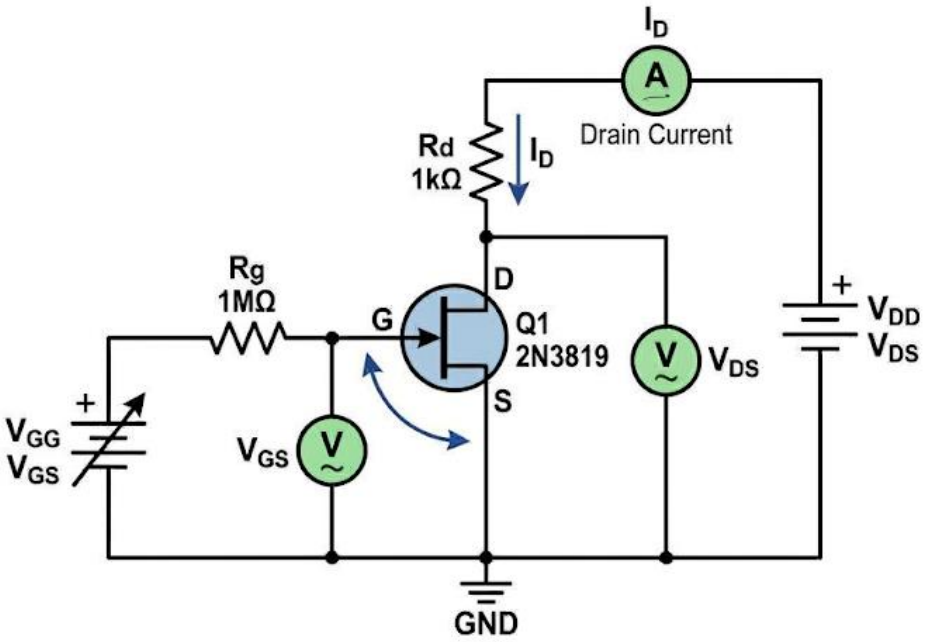
## Components and Tools Used

1. Simulation software : tinkercad , kicad ,or LTspice .
2. Active component : N-channel JEFT (eg. 2N3819 or equivalent)
3. Power & measurement : DC voltage sources ($V_{DD}$ and $V_{GG}$ ), multimeters, and ammeters to measure drain current ($I_D$) and drain-source voltage ($V_{DS}$) .

# Circuit Diagram

**Common-Source Configuration Setup**:

- The source terminal is connected as common to both the input and output loops.
- The gate terminal is connected to a negative DC voltage source ($V_{GG}$) to maintain reverse bias.
- The drain terminal is connected through a DC power supply ($V_{DS}$) to measure the output current ($I_D$).




**Circuit Diagram for Analyzing JFET Output Characteristics** ($V_{GS}$ vs $I_D$) - demonstrating JFET as a voltage-controlled current source with high input impedance.

# Explanation (CIRCUIT PARTS)

- **Circuit Operation**: The output characteristics curve plots the drain current ($I_D$) against the drain-source voltage ($V_{DS}$) at various constant gate-source voltages ($V_{GS}$). As $V_{DS}$ increases, $I_D$ initially increases linearly (ohmic region) and eventually flattens out into the saturation region.

- **(i) JFET is a voltage-controlled current source**:
- In the saturation region, the output drain current ($I_D$) is virtually independent of the drain-source voltage ($V_{DS}$) and is instead controlled entirely by the input gate-source voltage ($V_{GS}$). Changing $V_{GS}$ modulates the width of the internal depletion region, thereby regulating the channel width and controlling current flow.

- **(ii) JFET has high input impedance**:
- The gate-to-channel junction of the JFET is reverse-biased during normal operation. This reverse bias restricts any significant gate current from flowing, leading to an extremely high input resistance (often in the megaohm range), which gives the device its characteristic high input impedance.

# Simulations circuit

- **simulation Steps**:

    - Construct the common-source JFET circuit in Tinkercad or KiCAD.
    - Fix the gate-source voltage ($V_{GS}$) at a specific value (e.g.,..0V).
    - Sweep the drain-source voltage ($V_{DS}$) from 0V upwards and record the corresponding drain current ($I_D$).
    - Repeat the process for incremental negative values of $V_{GS}$ (e.g., -1V, -2V) to plot the complete family of output characteristic curves.

# Code

**Not Applicable (N/A)**: This task involves analog electronic circuit simulation and analysis, so no programming or microcontroller code is required.